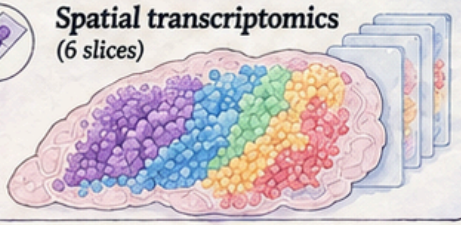
A computational framework for discovering a therapeutically targetable malignant cell state in glioblastoma

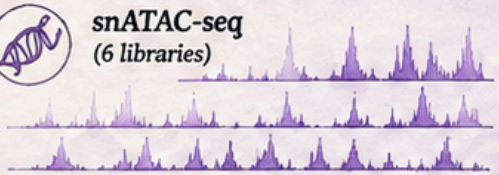
scRNA-seq
(35,823 cells)



Spatial transcriptomics
(6 slices)



snATAC-seq
(6 libraries)



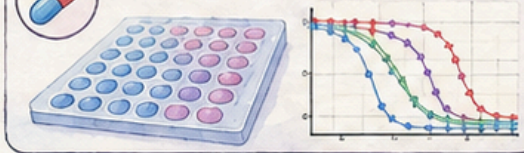
Bulk cohorts
(TCGA · CGGA · GSE · GLASS)



CPTAC proteomics
(99 samples, 11,141 proteins)

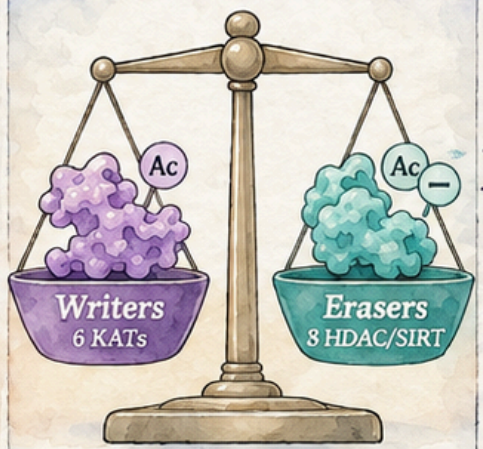


DepMap · GDSC2 screens



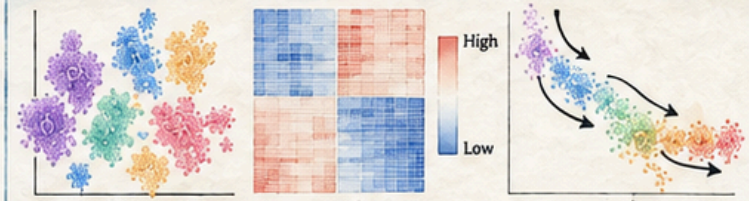
WEBs

Writer–Eraser Balance Score



LLS	LDTS	LHS
Low-balance state score	*Differentiation trajectory score*	*High-balance state score*

Discovery | *Malignant cell state landscape*



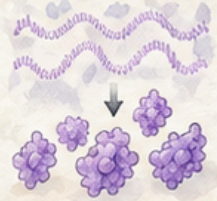
Cell-state landscape · *Hypoxia decoupling* · *Trajectory*

Mechanism | *Multi-omics mechanistic analysis*

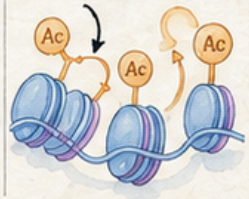
CPTAC spliceosome activation (107/111)



Post-transcriptional accumulation

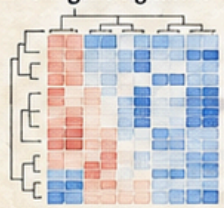


Acetylation coupling

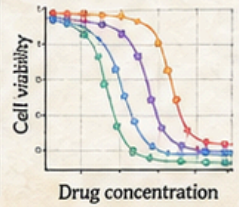


Translation | *Therapeutic and clinical relevance*

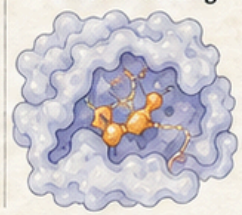
Six-gene signature



Pharmacogenomics



Molecular docking



KEY FINDING



Post-transcriptional spliceosome activation

Protein $\rho = +0.79$ *vs mRNA* $\rho = -0.03$

Discovery → Mechanism → Translation

All data, code and models are publicly available for reproducibility.